A Dynamic Co-Frequency Interference Analysis Model Based on Time-Elevation Interference Spectrum for NGSO Mega-Constellations

Zhaoyang Su, Kai Wang, Yuchen Cai, Lipeng Ning, Liu Liu, *Member, IEEE*, Tao Zhou, *Senior Member, IEEE*, and Bo Ai, *Fellow, IEEE*

***Abstract*—In recent years, satellite internet has been widely recognized as a key component of future integrated space-air-ground networks. With advancements in satellite miniaturization and launch technologies, mega-constellations have become a growing trend. The increasing number of satellites in constellations presents challenges for interference analysis. This paper proposes a novel interference analysis method based on time-elevation interference spectrum. The proposed method can provide a more comprehensive analysis for NGSO mega-constellations by considering the aggregated dynamic interference under the time-elevation domain. The interference characteristics under different orbital inclination, orbital plane numbers, orbital height, and ground station latitude are analyzed. Furthermore, the probability distributions of interference are derived based on the joint distribution of satellites and ground stations. The outage probabilities and throughput are also analyzed to measure the system's availability. Through the validation of STK and the Monte Carlo method, our method has high accuracy.**

***Index Terms*—Satellite communication, co-frequency interference, interference analysis, NGSO, LEO.**

I. Introduction

NOWADAYS, the development of terrestrial communication networks has advanced significantly, which can largely meet the daily communication needs of people in densely populated areas. However, in regions with harsh natural environments and oceans, which across approximately 70% of the earth's surface, terrestrial communication systems struggle to achieve full coverage. High deployment costs also make it difficult for impoverished areas to access the latest communication network services. Additionally, terrestrial networks have limited resilience against natural disasters. In contrast, satellite communication networks offer wide coverage and are unaffected by geographical constraints. Non-geostationary orbit (NGSO) constellations, particularly low Earth orbit (LEO) constellations, provide advantages such as lower deployment costs, lower communication latency, and higher communication bandwidth. Through the space-air-terrestrial integrated networks, global coverage and seamless connectivity can be realized. In recent years, satellite communication networks have rapidly advanced and are now recognized as a key component of the sixth-generation (6G) mobile communication system [1].

In recent years, due to technologies such as multiple satellites with one arrow and rocket reuse, NGSO constellations have been rapidly deployed. Representative NGSO constellations include Starlink, OneWeb, and TeleSat. The Starlink project plans to establish a constellation with 42000 satellites, OneWeb aims for a constellation of 2648 satellites, and TeleSat plans a constellation of 1671 satellites. As of June 2024, there are over 10000 active satellites in orbit globally, with more than 90% of them being LEO satellites. Thus, it can be seen that satellite communication has now entered the era of mega-constellations.

As the number of satellites in NGSO constellations increases, the scarcity of space frequency resources will inevitably become more pronounced [2]. Although different satellite operators can apply for independent frequency allocations, the scarcity of available spectrum often leads commercial NGSO constellations to operate within the same frequency bands. For example, Starlink and OneWeb both utilize the 10.7-12.7 GHz for satellite-to-user terminal links and the 12.75-13.25 GHz for user terminal-to-satellite links, as reported in FCC public filings [3], [4]. In addition, due to the difficulty of coordination among different satellite operators, frequency reuse among NGSO constellations within overlapping coverage areas is almost unavoidable. As a result, co-frequency interference (CFI) between NGSO constellations, as well as between NGSO and GSO constellations, will occur more frequently. Therefore, it is crucial to study CFI between satellite communication systems before deployment. Current researches mainly focus on interference between NGSO and GSO constellations [5], [6], [7], [8], [9], [10], where the GSO satellites remain stationary relative to the ground, and only the NGSO satellites exhibit relative motion concerning the Earth. These studies typically involve only a small number of satellites, with many focusing on interference

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The authors are with the School of Electronic and Information Engineering, Beijing Jiaotong University, Beijing 100044, China (e-mail: 21111042@bjtu.edu.cn; 20111045@bjtu.edu.cn; 23125016@bjtu.edu.cn; 27875108@qq.com; liuliu@bjtu.edu.cn; taozhou@bjtu.edu.cn; boai@bjtu.edu.cn).

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between a single NGSO and a single GSO satellite. However, with the recent development of mega-NGSO constellations, the complexity of CFI scenarios will significantly increase, exhibiting the following key characteristics [11]:

1) *Multiple interference sources*. The NGSO mega-constellations increase the probability that other constellation systems will appear within the field of view of ground stations. The aggregated interference from a large number of sidelobe signals can severely impact the normal reception at the receiver.
2) *Strong interference time-variability*. Unlike traditional NGSO-GSO interference scenarios, NGSO satellites are in continuous high-speed relative motion concerning the Earth. The geometric relationship between the satellites and the ground changes over time, leading to time-varying interference sources, interference targets, and interference links, which results in a highly dynamic interference scenario.
3) *Asymmetric interference*. Due to differences in the number of satellites and constellation configurations across various NGSO constellations, the interference intensity caused by different constellations varies. Consequently, the same interference analysis methods may not be universally applicable to all NGSO constellations.

Therefore, traditional interference analysis methods based on NGSO-GSO scenarios are no longer suitable for mega-constellations. It is essential to develop new interference analysis approaches.

A. Related Works

Currently, many researchers use different methods to determine the influence of CFI. Specifically, the existing research methods can be categorized into the following types.

1) *CFI analysis based on risk assessment*. This method establishes a risk assessment framework to analyze the severity of interference. It includes the types of interference that may occur in NGSO constellations, their probabilities of occurrence, and the severity of their harmful effects. By integrating these evaluation metrics, a multidimensional analysis is conducted to understand the impact of various factors on interference and system availability. Based on this approach, reference [12] analyzed the probability of harmful interference and its consequences for different NGSO constellations in the V-band. Reference [13] extended this work to the Ka-band and studied the impact of different interference mitigation strategies on system service metrics. However, this method is mainly used for qualitative analysis, and it is difficult to quantitatively describe the probability of CFI between constellations.
2) *CFI analysis based on numerical calculations*. For NGSO constellations, both the constellation configuration and communication parameters are predetermined. Therefore, the interference between constellations is only related to the positions of satellites and ground stations from different constellations. Once the position of a particular satellite is determined, the positions of other satellites in the same constellation can also be determined. Therefore, the numerical calculation method is considered an efficient way to analyze interference [14], [15]. Reference [14] derived the probability distribution function of NGSO satellites at various latitudes and longitudes. However, it introduced simplified assumptions for NGSO constellations, making it applicable only to constellations without repeat-orbit satellites, which limits its suitability for today's mega-constellations. Reference [16] proposed a snapshot-based probability distribution method to calculate interference between NGSO constellations, but it only considered a signal ground station. Reference [17] analyzed the CFI of the NGSO constellation to the GEO satellite IoT system by spatio-temporal slices and derived the outage probability, but it only considered the CFI between NGSO and GEO, and the influence of time and elevation was not researched. Authors in [18], [19], [20], and [21] used stochastic geometry-based methods to calculate satellite position coordinates and derive the probability distributions of interference, but true constellation configurations and the influence of satellite movement are neglected. In [22], authors modeled interference using a stochastic geometry framework to derive the coverage probability of multi-shelled LEO constellations, however, only the average interference was modeled and the dynamic interference is not considered.
3) *CFI analysis based on link angle*. This method starts by analyzing the relationship between the interference link angles formed by the communication links and the corresponding beam-pointing directions. It examines how different angles correlate with interference intensity and establishes a harmful interference threshold. By back-calculating the threshold angle for the link, harmful interference is considered to occur when the link angle is smaller than the threshold. Based on statistical results of link angles over a simulation period, the probability of harmful interference between two NGSO constellations is derived [23]. In [24], the authors derived the global distribution of harmful interference probability among different NGSO satellite constellations by researching the spatial isolation angle coordination interference index. Reference [25] proposed a calculation method of the angle threshold by using geometric configuration constraints. Authors in [26] deployed a platform to simulate and analyze NGSO CFI. However, this approach often classifies interference as harmful or non-harmful based on the angle threshold, which limits the accurate dynamic numerical analysis to the interference.

B. Motivations and Contributions

Most existing studies focus on specific scenarios, requiring a re-modeling when the constellation configuration or ground station distribution changes, thereby exhibiting limited generalizability. In addition, current research typically assumes that the interfered satellite system is static, without considering the

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mobility of both the interfering and interfered satellites. As a result, the spatial-temporal dynamics of inter-constellation interference remain underexplored. Moreover, most works rely on average interference power or interference probability metrics, which can not capture the temporal and spatial evolution of interference in detail.

Due to the temporal and spatial non-stationary characteristics of inter-constellation interference in NGSO mega-constellations, and the complex relative positions between interfering satellites, communication satellites, and interfered ground stations at different moments, traditional methods struggle to fully capture the variations in interference strength, which are highly dependent on specific link geometries. To address these limitations, this paper proposes a dynamic CFI interference analysis model based on the time-elevation interference spectrum (TEIS), which decouples time and spatial dimensions, thereby revealing interference evolution patterns from a macroscopic perspective and analyzing the impact of constellation configurations on interference dynamics. The main contributions of this work are as follows:

1) Parametric modeling of satellite constellations is developed to accurately determine the position of any satellite at any given time. By integrating stochastic modeling of ground station distributions, the proposed framework achieves strong generalizability across varying space-ground configurations and significantly improves interference analysis efficiency in different deployment scenarios.
2) A novel time-elevation analysis perspective is introduced for examining interference among NGSO mega-constellations. By decoupling the dynamics of the interfering constellation from the position of the interfered satellite, the model derives the TEIS, which enables intuitive identification of time periods and elevation angles where strong interference is likely to occur.
3) The paper investigates the interference characteristics under various space-ground configurations, revealing periodic interference patterns and the emergence of "interference hotspots." Furthermore, by utilizing the joint spatial distribution probabilities of satellites and ground stations, the probability density function (PDF) of the interference and the corresponding outage probability are derived. These findings can provide a theoretical foundation for the design of interference avoidance mechanisms among future NGSO mega-constellations.

The remainder of this paper is organized as follows: Section II presents the system model. Section III details the derivation of the proposed TEIS model. Section IV discusses simulation results. Section V concludes the paper.

II. System Model

This work investigates a downlink interference analysis model for NGSO mega-constellations, which are characterized by a massive number of satellites, high dynamic behavior, and rapidly changing interference links. This section introduces the model from three key perspectives: the downlink interference scenario, the propagation model, and the interference evaluation metrics.

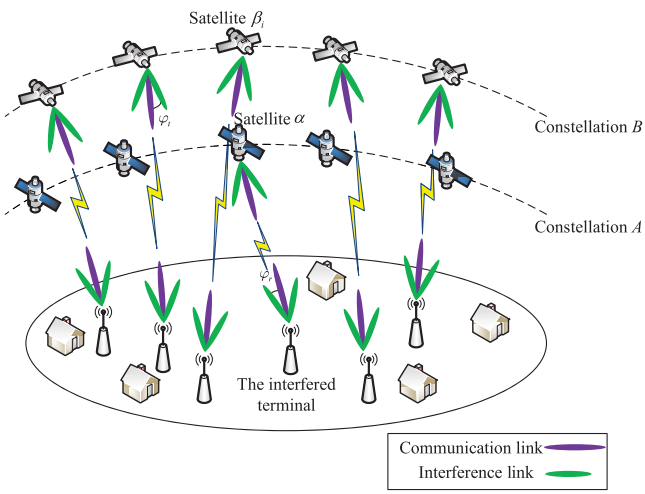

Fig. 1. Downlink interference scenario.

A. Downlink Interference Scenario

Fig. 1 illustrates the schematic of the downlink interference scenario, comprising two major components: the NGSO constellation and ground terminals. The NGSO constellation is further categorized into an interfering constellation and a communication constellation. In the figure, Constellation A serves as the communication constellation, while Constellation B acts as the interfering constellation. Ground terminals are classified into interfered terminal and interfering terminals. The interfered terminal establishes a communication link with a satellite from the communication constellation, and each interfering terminal communicates with a satellite from the interfering constellation using a maximum elevation angle criterion. Therefore, an Earth-fixed cell model is adopted, which is consistent with the operation mode of the fixed-satellite service (FSS) systems such as Starlink. A minimum elevation angle threshold $\theta_{\min}$ is utilized to determine the feasibility of link establishment. During the satellite's movement, its beam dynamically adjusts its pointing direction to continuously track the connected ground terminal. When the elevation angle decreases below the threshold, the current link is terminated, and the ground terminal establishes a new link with another high-elevation satellite. Each satellite that forms a link with an interfering terminal simultaneously generates an interference link toward the interfered terminal. Under this configuration, both the transmit antenna gain of the interfering satellite's sidelobe toward the interfered ground station and the receive antenna gain of the ground terminal's sidelobe toward the interfering satellite vary with time.

The interfered terminal may experience aggregated interference from multiple interfering satellites, leading to significant cumulative effects. To characterize the interference dynamics experienced by the interfered terminal under different temporal conditions and varying satellite associations, we consider a simplified scenario involving one interfered terminal, one communication satellite, and a set of interfering satellites

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from an interfering constellation. A circular region centered at the interfered terminal is defined to host the deployment of interfering terminals.

Since the range of elevation angles of satellites passing over a ground terminal varies with orbital inclination, an idealized configuration for the communication satellite is adopted to facilitate comprehensive interference characterization across all possible elevation angles. Specifically, the communication satellite is assumed to travel from south to north, passing directly over the zenith of the interfered terminal. This ideal trajectory enables consistent evaluation of interference effects under all potential link elevation conditions, while preserving the use of realistic constellation parameters for the interfering satellites.

B. Downlink Propagation Model

The downlink propagation model comprises three main components: the path loss model, the transmit antenna gain model, and the receive antenna gain model.

1) Path Loss Model: Unlike terrestrial cellular channels, satellite-to-ground channels are predominantly line-of-sight (LOS) and are mainly affected by free-space path loss, rain and cloud attenuation, and atmospheric absorption. The total path loss can be expressed as [27]:

$$PL = FSPL + PL_R + PL_A, \tag{1}$$

in which $FSPL$ denotes the free-space path loss, which is given by:

$$FSPL(d, f) = 32.45 + 20\log_{10}(d) + 20\log_{10}(f), \tag{2}$$

in which d is the propagation distance in kilometers and f is the carrier frequency in megahertz.

PL_R represents the attenuation due to rain and cloud when the satellite signal passes through the atmosphere. This component can be neglected below 6 GHz but becomes significant above 10 GHz or at low elevation angles around 1 GHz. It can be modeled as [28]:

$$PL_R = \gamma_R L_E, \tag{3}$$

in which γ_R is the specific attenuation in dB/km, determined based on the 0.01% annual rainfall rate and frequency-dependent coefficients from ITU-R P.838. L_E is the effective path length through the rain layer.

PL_A accounts for atmospheric absorption losses, which are non-negligible above 10 GHz. It can be modeled as [29]:

$$PL_A = \frac{PL_{zenith}}{\sin(\theta)}, \tag{4}$$

in which θ is the satellite elevation angle, and PL_{zenith} is the total zenith attenuation caused by oxygen and water vapor molecules, given by [29]:

$$PL_{zenith} = \gamma_o h_o + \gamma_w h_w, \tag{5}$$

where γ_o and γ_w are the specific attenuation coefficients in dB/km for oxygen and water vapor, respectively; h_o and h_w denote the equivalent vertical path lengths.

2) Transmit Antenna Gain Model: The transmit antenna gain is modeled according to ITU Recommendation S.1528, which defines a directional gain pattern suitable for satellite transmit antennas. The transmit antenna gain model can be expressed as [30]:

$$G(\varphi_t) = \begin{cases} G_{t,\max}, \varphi_t \le \varphi_b \\ G_{t,\max} - 3\left(\dfrac{\varphi_t}{\varphi_b}\right)^2, \varphi_b < \varphi_t < Y \\ G_{t,\max} + L_S - 25\log\left(\dfrac{\varphi_t}{Y}\right), Y < \varphi_t < Z \\ L_F, Z < \varphi_t < 180^\circ, \end{cases} \tag{6}$$

in which φ_t and φ_b are the off-axis angle and half of the 3 dB beam width at the maximum off-axis angle, L_S and L_F are the cross point of main beam and near sidelobe mask below peak and far-out side-lobe level. $G_{t,\max}$ is the maximum gain in the main lobe, which can be calculated as $G_{t,\max} = 20\log(D/\lambda)$. D represents the antenna diameter, and λ is the wavelength. For NGSO satellites, there are relationships as: $D/\lambda < 35$, $L_S = -6.75$, $Y = 1.5\varphi_b$, and $Z = Y \cdot 10^{0.04(G_{t,\max}+L_S-L_F)}$.

3) Receive Antenna Gain Model: The receive antenna gain at the ground terminal follows the pattern defined in ITU Recommendation S.465, which characterizes the gain pattern for earth station receiving antennas. The receive antenna gain model can be expressed as [31]:

$$G(\varphi_r) = \begin{cases} G_{r,\max}, \varphi_r < \varphi_m \\ 32 - 25\log(\varphi_r), \varphi_m \le \varphi_r \le 48^\circ \\ -10, \varphi_r > 48^\circ, \end{cases} \tag{7}$$

$$\varphi_m = \max\left[2,144\left(\frac{\mathrm{D}}{\lambda}\right)^{-1.09}\right]. \tag{8}$$

in which φ_r is the off-axis angle of the ground terminal.

C. Interference Evaluation Metrics

This section will derive the interference evaluation metrics based on the first principles, such as the basic definition of communication index and Shannon theorem. In this work, interference-to-noise ratio (INR) is utilized to analyze the interference of interfering satellites to the interfered terminal. The calculation method of INR is:

$$INR = \frac{P_T G_T(\varphi_t) G_R(\varphi_r) G_h}{kTB}, \tag{9}$$

in which P_T is the transmit power of satellites, $G_T(\varphi_t)$ is the gain of a satellite transmitting antenna when it is offset by an angle of φ_t from the boresight direction, $G_R(\varphi_r)$ is the gain of a ground terminal receiving antenna when it is offset by an angle of φ_r from the boresight direction, G_h represents the channel gain, k represents the boltzmann constant, T represents the receiver equivalent noise temperature.

To evaluate the service availability of the ground terminal under aggregate interference from a NGSO mega-constellation, the outage probability is employed. In the downlink scenario, the outage probability is defined as:

$$P_{outage} = \Pr\{SINR \le S_0\}, \tag{10}$$

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in which $SINR$ denotes the signal-to-interference-plus-noise ratio (SINR) at the interfered terminal, and S_0 is the minimum acceptable SINR threshold for reliable communication. An outage occurs when the $SINR$ falls below this threshold. The $SINR$ can be expressed as:

$$SINR = \frac{P_R}{I_{ag} + N_{noise}} = \frac{P_T G_{t,\max} G_{r,\max} G_h}{\sum_i I_{ag,i} + kTB}, \tag{11}$$

in which I_{ag} represents the aggregate interference power from all interfering satellites. S_0 can be expressed as follows according to Shannon's capacity theorem:

$$S_0 = 2^{C_0/B} - 1, \tag{12}$$

in which C_0 is the minimum data rate acquired for the interfered system. Therefore, the downlink outage probability for the interfered terminal can be finally expressed as:

$$P_{outage} = \Pr\left\{\sum_i I_{ag,i} \geq \frac{P_T G_{t,\max} G_{r,\max} G_h}{2^{C_0/B} - 1} - kTB\right\}. \tag{13}$$

In addition, throughput is also an important metric to evaluate the system performance, which can be expressed as following based on the outage probability:

$$THU = B\left(1 - P_{outage}\right)\log_2\left(1 + SINR\right). \tag{14}$$

III. Downlink Interference Analysis Based on TEIS

In this section, a downlink interference analysis based on the proposed TEIS is conducted. The mathematical analysis mainly consists of six parts: (1) constellation parameter derivation; (2) satellite position estimation; (3) constellation period calculation; (4) ground station position derivation; (5) interfering satellite distribution characterization; and (6) interference probability derivation.

Compared with conventional terrestrial cellular systems, NGSO satellite communication exhibits two distinct characteristics: high mobility and the existence of elevation angles, which correspond to temporal and spatial non-stationarity, respectively. Therefore, this section starts from the orbital parameters of satellites and derives the instantaneous positions of both satellites and ground terminals, followed by a fast identification of interfering satellites. Based on these relationships, the joint time–elevation domain expression of the TEIS is formulated to reveal the evolution patterns of aggregated interference on the spatio-temporal plane. Furthermore, the probability distribution of downlink interference is also derived to further investigate the statistical characteristics of interference behavior.

A. NGSO Constellation Structure and Satellite Position Computation

1) Constellation Structure: In this study, the widely adopted Walker constellation is employed to model the interfering satellite system. The interfering constellation is defined as $N_I/P_I/F_I : h_I, i_I$, where N_I is the total number of satellites, P_I is the number of orbital planes, F_I is the phasing factor that determines the phase difference between satellites in adjacent planes, h_I is the orbital altitude, and i_I is the orbital inclination. Since the constellation configuration is fixed, the positions of all satellites can be uniquely determined once a reference satellite is specified [15]. Thus, the concept of a "reference satellite" is introduced to determine the distribution of all satellites. The diagram of constellation configuration is shown in Fig. 2.

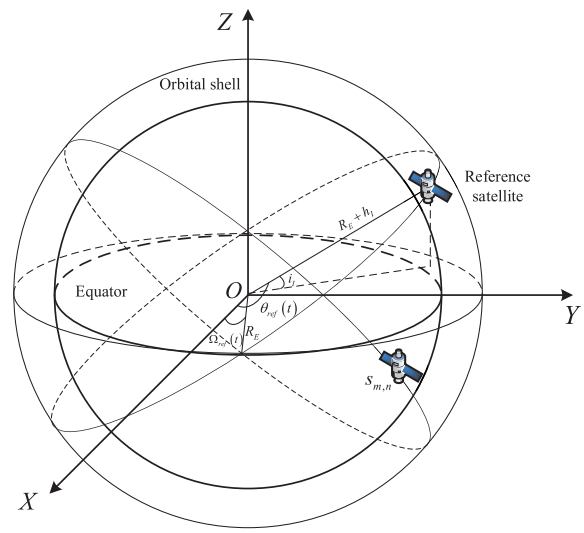

Fig. 2. Diagram of constellation configuration.

An Earth-centered Earth-fixed (ECEF) coordinate system is established, with the origin at the Earth's center. The Z-axis aligns with the Earth's rotation axis, pointing to the North Pole, the X-axis points to the intersection of the prime meridian and the equator, and the Y-axis points to the intersection of 90° E longitude and the equator. Assuming satellites follow ideal circular orbits, the right ascension of the ascending node (RAAN) of the reference satellite is set to zero, and the mean anomaly is also zero at the initial moment. At time t, the mean anomaly of the reference satellite is denoted as $\theta_{ref}(t)$. The RAAN and mean anomaly of the $n-th$ satellite on the $m-th$ orbital plane can be expressed as:

$$\Omega_{m,n}(t) = \Omega_{ref}(t) + \frac{2\pi}{P_I} m, \tag{15}$$

$$\theta_{m,n}(t) = \theta_{ref}(t) + 2\pi\left(\frac{F_I}{N_I} m + \frac{P_I}{N_I} n\right), \tag{16}$$

$$\theta_{ref}(t) = \omega_s t = \sqrt{\frac{GM}{{r_s}^3}}\, t, \tag{17}$$

in which r_s is the orbital radius satisfying $r_s = R_E + h_I$, R_E is the Earth's radius, G is the gravitational constant, and M is the Earth's mass. Taking into account Earth's rotation, the RAAN of the reference satellite at time t becomes:

$$\Omega_{ref}(t) = \frac{-2\pi t}{T_E}, \tag{18}$$

where T_E is the length of a sidereal day, approximately 23 hours, 56 minutes, and 4 seconds.

2) Calculation of Satellite Position: To determine the interference, the position of satellites should be calculated. For the

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$n-th$ satellite on the $m-th$ orbital plane, which is denoted as $s_{m,n}$, the position in the ECEF coordinate system can be expressed as:

$$s_{m,n}^{ECEF}(t) = r_s \begin{bmatrix} \cos\phi_{m,n}(t)\cos\Lambda_{m,n}(t) \\ \cos\phi_{m,n}(t)\sin\Lambda_{m,n}(t) \\ \sin\phi_{m,n}(t) \end{bmatrix}, \tag{19}$$

in which $\Lambda_{m,n}(t)$ and $\phi_{m,n}(t)$ represent the longitude and latitude of the subsatellite point of $s_{m,n}$. In order to derive the $\Lambda_{m,n}(t)$ and $\phi_{m,n}(t)$ from the satellite's orbital parameters, coordinate system transformation is necessary, i.e., the conversion from perifocal coordinate (PQW) system to Earth-centered inertial (ECI) coordinate system.

In PQW system, the position of $s_{m,n}$ can be expressed as:

$$s_{m,n}^{PQW}(t) = \left[r_s\cos\theta_{m,n}(t) \;\; r_s\sin\theta_{m,n}(t) \;\; 0 \right]^T. \tag{20}$$

Define the transformation matrix from PQW to the ECI coordinate system as $\mathbf{R}_{PQW\to ECI}$. For circular orbits, $\mathbf{R}_{PQW\to ECI}$ can simplify to:

$$\mathbf{R}_{PQW\to ECI} = \begin{bmatrix} \cos\Omega_{m,n}(t) & -\sin\Omega_{m,n}(t)\cos i_I & \sin\Omega_{m,n}(t)\sin i_I \\ \sin\Omega_{m,n}(t) & \cos\Omega_{m,n}(t)\cos i_I & -\cos\Omega_{m,n}(t)\sin i_I \\ 0 & \sin i_I & \cos i_I \end{bmatrix}. \tag{21}$$

The coordinate of $s_{m,n}$ in ECI coordinate system can be expressed as:

$$s_{m,n}^{ECI}(t) = \mathbf{R}_{PQW\to ECI}\cdot s_{m,n}^{PQW}(t). \tag{22}$$

Therefore, the longitude and latitude of $s_{m,n}$ can then be expressed as:

$$\Lambda_{m,n}(t) = \arctan\left(\frac{y_{m,n}^{ECI}(t)}{x_{m,n}^{ECI}(t)}\right), \tag{23}$$

$$\phi_{m,n}(t) = \arcsin\left(\frac{z_{m,n}^{ECI}(t)}{r_s}\right), \tag{24}$$

in which $x_{m,n}^{ECI}(t)$, $y_{m,n}^{ECI}(t)$, and $z_{m,n}^{ECI}(t)$ are the X, Y, Z coordinates of $s_{m,n}$ in ECI coordinate system, respectively.

Hence, the position of any satellite in a Walker constellation at any time is obtained.

3) Determination of Constellation Period: Since the Walker constellation has a fixed geometric structure, the relative positions among satellites exhibit periodic patterns. Consequently, the number and distribution of interfering satellites above an interfered ground terminal also show periodic variation, which defines the constellation period. Identifying this period is essential for setting appropriate analysis durations when analyzing interference evolution over time.

Assume the initial time is t_0, and consider a satellite $s_{m,n}$ above the interfered ground terminal. Due to satellite motion and Earth's rotation, if after time T a satellite from a different orbital plane passes over the same sub-satellite point (SSP), then $T = T_{period}$. Given that the RAAN difference between adjacent planes is $\Delta\Omega = 2\pi/P_I$, the ground track of the $(m+1)-th$ orbital plane will coincide with that of the $m-th$ orbital plane at time t_0 after $\Delta t_1 = \Delta\Omega\cdot T_E/2\pi$, which is referred to as the RAAN regression period. When the phase factor of the Walker constellation is F_I, the phase offset between satellites $s_{m,n}$ and $s_{m+1,n}$ can be expressed as $\Delta M = 2\pi F_I/N_I$. Consequently, after $\Delta t_2 = (2\pi - \Delta M)/\omega_s$, the phase of $s_{m+1,n}$ will return to the same value as that of $s_{m,n}$ at time t_0. Therefore, Δt_2 is defined as the phase difference regression period.

To ensure the complete geometric repetition of the constellation configuration, both the RAAN regression and the phase difference regression conditions must be satisfied simultaneously. Therefore, the constellation repetition period T_{period} can be expressed as:

$$T_{period} = \frac{\Delta\Omega\cdot T_E(2\pi - \Delta M)}{2\pi\omega_s \gcd(\Delta\Omega\cdot T_E/2\pi, (2\pi-\Delta M)/\omega_s)}, \tag{25}$$

in which $\gcd(,)$ denotes the greatest common divisor. It is important to note that T_{period} represents the full geometric repetition period of the entire constellation, which serves as the upper bound for the analysis time. For an individual interfering satellite, a complete repetition of the constellation is required for the interference intensity to return to its original state.

However, it should be noted that in the analysis of aggregate downlink interference, the interfered ground terminal is simultaneously affected by multiple satellites. As a result, it is not necessary for each individual satellite to complete a full geometric cycle for the interference pattern to repeat. Therefore, the actual interference recurrence period in the downlink is typically shorter than T_{period}.

B. Ground Terminal Location Distribution Calculation

Assume the interfered ground terminal is denoted by GS_{intd}, with geographic coordinates $(\Lambda_{gs}, \phi_{gs})$. Its position in the ECEF coordinate system can be expressed as:

$$GS_{intd} = R_E \begin{bmatrix} \cos\phi_{gs}\cos\Lambda_{gs} \\ \cos\phi_{gs}\sin\Lambda_{gs} \\ \sin\phi_{gs} \end{bmatrix}. \tag{26}$$

Since the Poisson Point Process (PPP) is commonly used to model the spatial distribution of large-scale terrestrial base stations [32], this paper also adopts the PPP to construct the spatial distribution model for interfering ground terminals. Taking the interfered ground terminal as the origin, a local east-north-up (ENU) coordinate system is established. For an interfering ground terminal $GS_{intg,ij}$ located at a distance r_i from the origin and at an azimuth angle θ_j, its position in the ENU coordinate system is given by:

$$GS_{intg,ij}^{ENU} = [r_i\cos\theta_j, r_i\sin\theta_j, 0]^T. \tag{27}$$

Define the transformation matrix from the ENU coordinate system to the ECEF coordinate system as $\mathbf{R}_{ENU\to ECEF}$, which can be expressed as:

$$\mathbf{R}_{ENU\to ECEF} = \begin{bmatrix} -\sin\Lambda_{gs} & -\sin\phi_{gs}\cos\Lambda_{gs} & \cos\phi_{gs}\cos\Lambda_{gs} \\ \cos\Lambda_{gs} & -\sin\phi_{gs}\sin\Lambda_{gs} & \cos\phi_{gs}\sin\Lambda_{gs} \\ 0 & \cos\phi_{gs} & \sin\phi_{gs} \end{bmatrix}. \tag{28}$$

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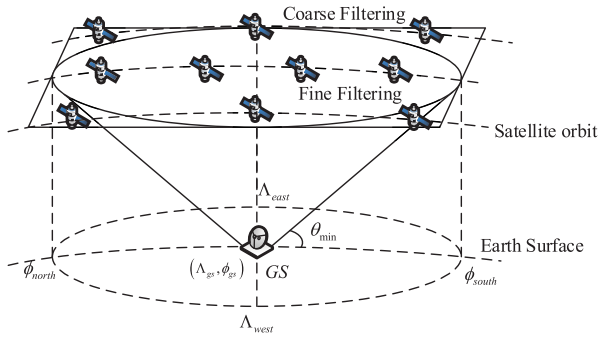

Fig. 3. Diagram of coarse filtering and fine filtering.

Therefore, the position of any interfering ground terminal $GS_{intg,ij}$ in the ECEF coordinate system can be expressed as:

$$\begin{aligned} GS_{intg,ij}^{ECEF} &= \mathbf{R}_{ENU\to ECEF} \cdot GS_{intg,ij}^{ENU} \\ &= r_i \begin{bmatrix} -\sin\Lambda_{gs}\cos\theta_j - \sin\phi_{gs}\cos\Lambda_{gs}\sin\theta_j \\ \cos\Lambda_{gs}\cos\theta_j - \sin\phi_{gs}\sin\Lambda_{gs}\sin\theta_j \\ \cos\phi_{gs}\sin\theta_j \end{bmatrix}. \end{aligned} \tag{29}$$

C. Dynamic Characteristics Analysis of Interfering Satellites

Due to the number of satellites being enormous in NGSO mega-constellations, efficiently identifying the quantity and location of interfering satellites is crucial for interference analysis. In this section, a two-stage method combining coarse filtering and fine filtering is proposed to determine the interfering satellites. The diagram of coarse filtering and fine filtering is shown in Fig. 3.

1) Coarse Filtering: In the previous derivation, the SSP latitude and longitude of each satellite at any given time can be calculated. Coarse filtering selects satellites whose SSPs fall within the visible region of the interfered ground terminal by determining the geographic bounds of this region. Assuming the interfered ground terminal is located at latitude ϕ_{gs} and the minimum elevation angle for establishing a link is $\theta_{\min}$, the minimum and maximum latitudes of the SSPs of satellites that are within the visible region can be given by:

$$\phi_{south} = \phi_{gs} - \arccos\left(\frac{R_E\cos\theta_{\min}}{R_E + h_I}\right) + \theta_{\min}, \tag{30}$$

$$\phi_{north} = \phi_{gs} + \arccos\left(\frac{R_E\cos\theta_{\min}}{R_E + h_I}\right) - \theta_{\min}. \tag{31}$$

Given that the latitude of the interfered ground terminal is Λ_{gs}, and assuming a satellite's SSP has geographic coordinates $(\Lambda_{m,n}(t), \phi_{m,n}(t))$, a spherical triangle is formed on the Earth's surface among the ground terminal, the satellite's SSP, and the Earth's pole. According to the spherical law of cosines, we have:

$$\cos\psi = \sin\phi_{m,n}(t)\sin\phi_{gs} + \cos\phi_{m,n}(t)\cos\phi_{gs}\cos\Delta\Lambda, \tag{32}$$

where $\Delta\Lambda$ is the longitude difference between the SSP and the ground terminal, and ψ is the central angle corresponding to the arc distance between the SSP and the ground terminal on the Earth's surface, satisfying:

$$\psi = \arccos\left(\frac{R_E\cos\theta}{R_E + h_I}\right) - \theta, \tag{33}$$

where θ is the satellite's elevation angle at that moment. Therefore, the minimum and maximum longitudes of visible SSPs can be expressed as:

$$\Lambda_{west} = \Lambda_{gs} - \arccos\left(\frac{\sin^2\phi_{gs} - \cos\psi_{\max}}{\cos^2\phi_{gs}}\right), \tag{34}$$

$$\Lambda_{east} = \Lambda_{gs} + \arccos\left(\frac{\sin^2\phi_{gs} - \cos\psi_{\max}}{\cos^2\phi_{gs}}\right), \tag{35}$$

where $\psi_{\max}$ is the value when $\theta = \theta_{\min}$. By checking whether each satellite's SSP lies within the latitude range $(\Lambda_{west}, \Lambda_{east})$ and longitude range $(\phi_{south}, \phi_{north})$, a preliminary set of potentially interfering satellites can be identified via coarse filtering.

2) Fine Filtering: Since the coarse filtering method yields a rectangular visible region, whereas the actual visible region resembles a circular area, the coarse filter may include non-interfering satellites. Fine filtering is therefore required.

According to Eq. (26), if there is a satellite s_{gs} directly above the zenith of the interfered ground terminal, its position in the ECEF coordinate system is:

$$s_{gs} = (R_E + h_I)\begin{bmatrix} \cos\phi_{gs}\cos\Lambda_{gs} \\ \cos\phi_{gs}\sin\Lambda_{gs} \\ \sin\phi_{gs} \end{bmatrix}. \tag{36}$$

Thus, for any satellite $s_{m,n}$ to be excluded from the set of interferers at the current time, the following condition must be satisfied:

$$\arccos\left(\frac{\langle s_{m,n} - GS, s_{gs} - GS\rangle}{\|s_{m,n} - GS\|\,\|s_{gs} - GS\|}\right) > \frac{\pi}{2} - \theta_{\min}. \tag{37}$$

By combining coarse and fine filtering, the set of satellites capable of interfering with the ground terminal at each moment can be accurately and efficiently determined. Denote the set of all such interfering satellites as $\left[s_1^I(t), s_2^I(t), \cdots, s_N^I(t)\right]$.

D. Expression of TEIS

In the preceding sections, the positions of both the interfered ground terminal and the interfering ground terminals have been obtained, as well as the number and positions of interfering satellites at any given time. Assuming the elevation angle of the communication satellite is θ_{com}, and its position is denoted by $s(\theta_{com})$, then according to Eq. (9), the downlink TEIS between NGSO mega-constellations can be expressed as:

$$\begin{aligned} INR(t, \theta_{com}|\Theta) &= \frac{P_T G_T}{kTB} \\ &\cdot \sum_{u=1}^{N} (\varphi_{t,u}(t|\Theta)) G_R(\varphi_{r,u}(t, \theta_{com}|\Theta)) G_{h,u}(t, \theta_{com}|\Theta), \end{aligned} \tag{38}$$

in which $\Theta = \{N_I, P_I, F_I, i_I, h_I\}$, N denotes the number of interfering satellites within the visible region of the interfered ground terminal at time t. $\varphi_{t,i}(t|\Theta)$ is the off-axis gain at the transmitting end of the $u-th$ interfering satellite's interference link, and can be expressed as:

$$\begin{aligned} &\varphi_{t,u}(t|\Theta) = \\ &\arccos\left(\frac{\langle GS_{intd} - s_u^I(t), GS_{intg,ij} - s_u^I(t)\rangle}{\|GS_{intd} - s_u^I(t)\|\,\|GS_{intg,ij} - s_u^I(t)\|}\right). \end{aligned} \tag{39}$$

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$\varphi_{r,u}(t,\theta_{com}|\Theta)$ is the off-axis gain at the receiving end of the $u-th$ interfering satellite's interference link, and can be expressed as:

$$\varphi_{r,u}(t,\theta_{com}|\Theta) = \arccos\left(\frac{\left\langle GS_{intd}-s_u^I(t), GS_{intd}-s(\theta_{com})\right\rangle}{\|GS_{intd}-s_u^I(t)\|\,\|GS_{intd}-s(\theta_{com})\|}\right). \quad (40)$$

$G_{h,u}(t,\theta_{com}|\Theta)$ is the path gain of the interference link from the $u-th$ satellite to the interfered ground terminal at time t, and can be expressed as:

$$G_{h,u}(t,\theta_{com}|\Theta) = 10^{PL(t,\theta_{com})/10}. \quad (41)$$

Since the TEIS expression jointly incorporates both time and elevation, it differs from conventional interference analysis methods that are typically conducted under static conditions. This formulation reveals how the aggregated interference evolves with respect to two key characteristics of satellite communication, i.e., temporal variation and elevation dynamics, thus enabling joint observation of interference in the time–elevation domain. Moreover, as the expression is fully derived from orbital parameters, it can be readily applied to any NGSO constellation, demonstrating strong scalability and general applicability.

E. Statistical Analysis of Downlink Interference

In addition to characterizing the time–elevation angle domain interference variation, it is also essential to analyze the statistical distribution of the interference. This section derives the conditional elevation angle-based INR probability distribution by jointly considering the probability of satellite occurrence and the distribution of ground stations.

Since the positions of satellites in the interfering constellation are uniquely determined by the reference satellite, the spatial distribution of all satellites in the constellation at any given moment can be uniquely characterized by the occurrence probability of the reference satellite. Suppose the total constellation analysis time is divided into $[0,\Delta t,2\Delta t,\cdots(n-1)\Delta t,n\Delta t,\cdots T_{period}]$, then the probability of the reference satellite appearing within the sky coverage in the $[(n-1)\Delta t,n\Delta t]$ time interval can be expressed as [32]:

$$p = \frac{A_c}{2\pi^2\sin\left(\arccos\frac{\cos i_I}{\cos L}\right)\cos L}, \quad (42)$$

where A_c denotes the spherical solid angle of the sky region covered by the reference satellite during the interval, and L is the central latitude of this sky region. A_c can be further calculated as:

$$A_c = \frac{\pi}{4}(\beta_2-\beta_1)(\varepsilon_2-\varepsilon_1), \quad (43)$$

where $\beta_2-\beta_1$ and $\varepsilon_2-\varepsilon_1$ are the geocentric angular spans of the latitude and longitude ranges, respectively. When Δt is sufficiently small, the coverage region can be approximated as a spherical rectangle with $(\Lambda_{ref}(n\Delta t),\phi_{ref}(n\Delta t))$ and $(\Lambda_{ref}((n-1)\Delta t),\phi_{ref}((n-1)\Delta t))$. Then A_c and L can be expressed as:

$$A_c = \frac{\pi}{4}(\Lambda_{ref}(n\Delta t)-\Lambda_{ref}((n-1)\Delta t)) \times(\phi_{ref}(n\Delta t)-\phi_{ref}((n-1)\Delta t)), \quad (44)$$

$$L = \frac{\phi_{ref}(n\Delta t)+\phi_{ref}((n-1)\Delta t)}{2}. \quad (45)$$

Therefore, the probability associated with the constellation's spatial configuration during the time interval $[(n-1)\Delta t,n\Delta t]$ can be written as:

$$p_{(n-1)\Delta t}^{n\Delta t} = \frac{\Lambda_{ref}(n\Delta t)-\Lambda_{ref}((n-1)\Delta t)}{8\pi\sin\left(\arccos\frac{\cos i_I}{\cos L}\right)} \times\frac{\phi_{ref}(n\Delta t)-\phi_{ref}((n-1)\Delta t)}{\cos\left(\frac{\phi_{ref}(n\Delta t)+\phi_{ref}((n-1)\Delta t)}{2}\right)}. \quad (46)$$

Since interfering ground terminals follow a PPP, the probability when the number of interfering ground terminals is N_{gs} can be expressed by:

$$p(N_{gs}) = \frac{(\pi\eta R^2)^{-N_{gs}}e^{-\pi\eta R^2}}{N_{gs}!}, \quad (47)$$

where η is the strength of the Poisson distribution, i.e., the average number of ground terminals per unit area. The value of η can be determined according to [33]. Due to the independence between satellite and ground terminal distributions, the probability of INR for a satellite–ground configuration with N_{gs} interfering ground terminals in the time interval $[(n-1)\Delta t,n\Delta t]$ can expressed as:

$$p_{INR}((n-1)\Delta t,n\Delta t,N_{gs}|\theta_{com}) = p_{(n-1)\Delta t}^{n\Delta t}p(N_{gs}), \quad (48)$$

Based on the expression of the PDF of INR, the outage probability can be rewritten as following according to Eq. (13):

$$\begin{aligned} &P_{outage}((n-1)\Delta t,n\Delta t,N_{gs}|\theta_{com}) \\ &= \Pr\left\{\sum_i INR_{ag,i} \geq \frac{SNR((n-1)\Delta t,n\Delta t,N_{gs}|\theta_{com})}{2^{C_0/B}-1}-1\right\} \\ &= 1-F_{INR}\left(\frac{SNR((n-1)\Delta t,n\Delta t,N_{gs}|\theta_{com})}{2^{C_0/B}-1}-1\right), \end{aligned} \quad (49)$$

in which $SNR(\cdot)$ is the signal-to-noise ratio (SNR) of the interfered terminal in a specific state, and $F_{INR}(\cdot)$ represents the cumulative distribution function (CDF) of INR. Assuming that CT_{outage} is the criteria of the outage probability, then the condition of $P_{outage}((n-1)\Delta t,n\Delta t,N_{gs}|\theta_{com}) < CT_{outage}$ can be expressed as:

$$F_{INR}\left(\frac{SNR((n-1)\Delta t,n\Delta t,N_{gs}|\theta_{com})}{2^{C_0/B}-1}-1\right) > 1-CT_{outage}. \quad (50)$$

Therefore, to satisfy the system availability, some system parameters such as transmit power and antenna gain can be

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TABLE I
SIMULATION PARAMETERS SETTINGS

Parameters	Values
Carrier frequency	20 GHz
Orbit height	550 km, 1200 km
Orbital inclination	80°, 50°
Number of satellites	72, 36
Maximum number of ground terminals	100
Satellite transmit power [28]	38 dBW
Bandwidth [28]	400 MHz
Peak gain of satellite transmitting antenna	30.5 dBi
Peak gain of ground terminal receiving antenna	32 dBi
Receiver system noise temperature	150 K
Elevation angle threshold	25°

adjusted based on the derived relationship, so as to maintain the outage probability below a target level.

In summary, a general INR probability expression under arbitrary communication satellite elevation angles and any satellite–ground configuration is derived in this section. By traversing different configurations under varying elevation angles, the conditional INR distribution with respect to elevation angle can be obtained. According to the expression, the outage probability can be directly derived and the evaluation of system service capability can be enabled at a given time and elevation angle. In addition, the INR PDF can also reveal the distribution of interference intensity under different elevation angles, which helps identify the angular regions with stronger interference and provides a priori information for interference avoidance.

IV. SIMULATION RESULTS

This section shows the simulation results of the proposed CFI analysis method based on MATLAB and systems tool kit (STK). STK is a professional software developed by Analytical Graphics Inc. (AGI) for simulating and analyzing satellite constellations and communication links. It is widely used as a benchmark tool in satellite communication research to validate orbital geometry and visibility statistics. The simulation results depict the interference characteristics under different constellation configurations, including the TEIS results, the PDF of INR, the outage probabilities, and throughput. The simulation parameters are shown in Tab. I [34].

In Tab. I, the orbital inclinations are both correspond to the interfering constellations. For communication constellation, as mentioned in system model section, a simplified communication constellation model is adopted, i.e., only one communication satellite is considered, which passes from the south to the north over the zenith of the ground terminal, thereby covering all elevation angles. Consequently, the orbital inclination of the communication satellite can be regarded as 90°. For the other parameters listed in Tab. I, the same values are used for both the communication constellation and the interfering constellation.

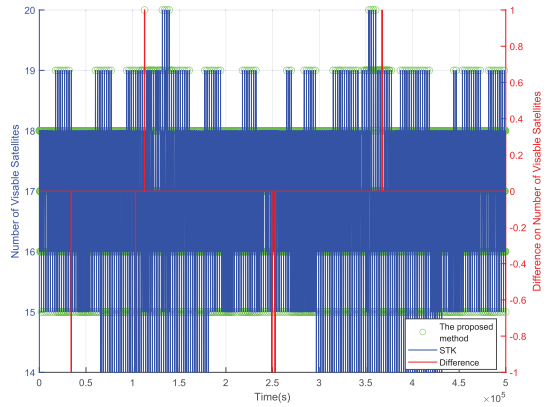

Fig. 4. The number of visible satellites.

TABLE II
THE STATISTICAL COMPARISON OF THE NUMBER OF VISIBLE SATELLITES

Method	Mean Value	Variance
The method proposed in this paper	16.7764	1.1539
STK	16.7768	1.1536

In the simulation, LEO constellations are selected as the focus because they represent the most interference-intensive scenario among NGSO systems, owing to their high velocity and rapidly changing satellite–ground topology. Nevertheless, the proposed TEIS-based analytical model can be extended to other constellations by adjusting the constellation parameters. Furthermore, the proposed model can also accommodate any number of interfering constellations by simultaneously applying different constellation parameters and including all satellites in the interference computation. However, considering that introducing multiple interfering constellations would significantly increase the computational complexity and simulation cost, and the aggregated interference from heterogeneous constellations with diverse parameters would be difficult to analyze explicitly, only one interfering constellation is simulated to balance complexity and representativeness.

Fig. 4 presents a comparison of the number of satellites within the visible region of the interfered ground station over time, obtained using both STK and the proposed method, and the difference in the satellites number the two methods is also depicted. It should be noted that the proposed method refers to the results obtained through coarse filtering and fine filtering for identifying the number of interfering satellites. As shown in the figure, the results derived from the proposed method exhibit good agreement with those from STK, thereby validating the correctness of the proposed approach and the subsequent simulation results. Additionally, it can be observed that the variation in the number of visible satellites displays a clear periodicity, which is attributed to the periodic nature of the Walker constellation. The mean and variance of the results obtained from the two methods are summarized in Tab. II.

Fig. 5 presents the simulated TEIS for INR when the latitude of the interfered ground terminal is located in Beijing, i.e.,

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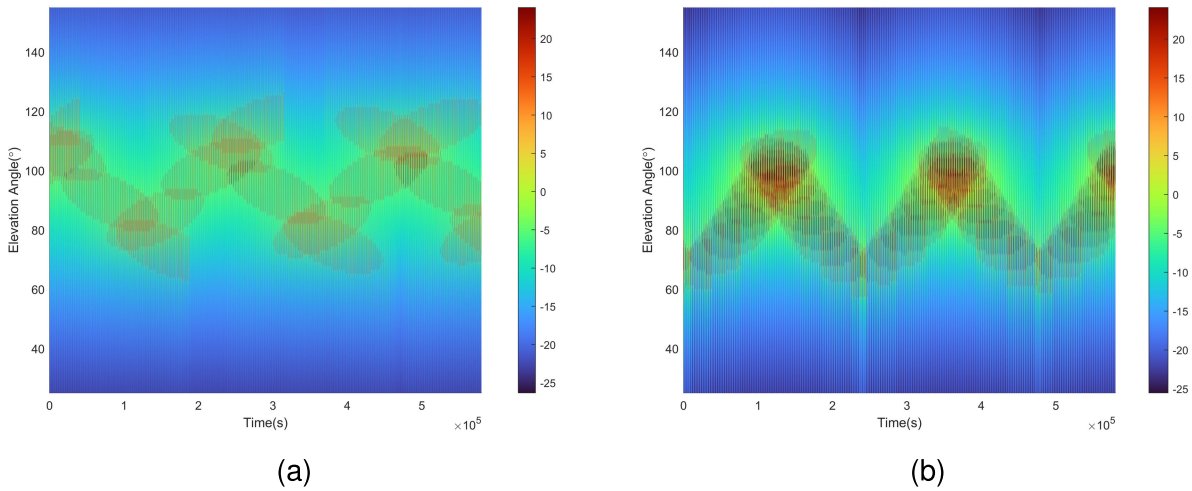

Fig. 5. The simulated TEIS for INR:(a) inclination=80°,(b) inclination=50°.

39° N and 116° E. The elevation angle refers to the elevation of the communication satellite. It is important to note that for ease of visualization, the elevation angle in the figure is defined as the angle between the communication link and the southern horizon. Therefore, when the elevation angle shown in the figure exceeds 90°, the actual elevation angle begins to decrease.

Fig. 5 (a) illustrates the results for an orbital inclination of 80°. It can be observed that the elevation angle of the communication satellite under strong interference exhibits a periodic change trend over time, which is caused by the periodic motion of the Walker constellation. For a fixed interfered ground terminal, the interfered elevation angle shows an increasing-then-decreasing pattern because the interfering satellites approach the station either from south to north or from north to south. When interference is predominantly caused by satellites flying from south to north, the interfered elevation angle tends to increase; conversely, when the interference mainly originates from satellites flying from north to south, the interfered elevation angle tends to decrease. Dense interference events often occur at the intersection of these two types of satellite motion.

In addition, multiple interference aggregation regions, referred to as "interference hotspots", are clearly observed in the time–elevation angle domain. These hotspots overlap and appear as elliptical regions in the figure. This phenomenon arises because, when the interfered ground terminal is located at 39° N and the orbital inclination is 80°, the number of visible satellites above the terminal is relatively limited. Since the interfering ground terminals follow a maximum-elevation-angle link establishment strategy, different interfering ground terminals may establish links with different satellites. Over a given period, the satellites linked with one terminal tend to belong to the same or adjacent orbital planes, forming clustered satellite groups and leading to multiple interference hotspot regions. In other words, these hotspots result from the aggregated interference effect of multiple interfering ground terminals and multiple interfering satellites on a single interfered ground terminal.

Fig. 5 (b) shows the results for an orbital inclination of 50°. The overall trend is similar to that in Fig. 5 (a). However, the strong interference regions in the time–elevation angle domain appear more concentrated, and the interference hotspots overlap more tightly. This is because with the inclination is 50°, the constellation provides enhanced coverage for mid-latitude ground terminals, increasing the number of visible satellites above the terminal. The satellites linked by different ground terminals change more rapidly, leading to denser overlapping interference hotspots. Therefore, orbital inclination significantly affects the concentration of interference in the time–elevation angle domain.

Fig. 6 presents the simulated TEIS for different ground terminal latitudes, with the orbital inclination fixed at 80°. In Fig. 6 (a), it can be observed that when the ground terminal is located at a lower latitude of 10°, the overall elevation range at which the interfered ground terminal experiences heavy interference is reduced. This is due to the generally lower satellite elevation angles observable from low-latitude ground terminals. Additionally, compared to Fig. 5 (a), the "interference hotspots" appear more sparsely distributed. This is because fewer satellites are visible to low-latitude ground stations, resulting in greater divergence in the satellites linked by different interfering terminals.

In Fig. 6 (b), where the ground terminal is located at a high latitude of 70°, the elevation angles are more tightly around 90°, and the "interference hotspots" overlap more densely. This occurs because the ground terminal is positioned near the edge of the satellite orbit coverage, where the satellite density is higher. Consequently, the satellites linked by different ground terminals are more closely spaced. Furthermore, due to the high orbital inclination, the satellites appear at higher elevation angles, causing the linked satellites to concentrate near the zenith.

In summary, ground terminal latitude significantly affects both the range and density of interference in the time–elevation

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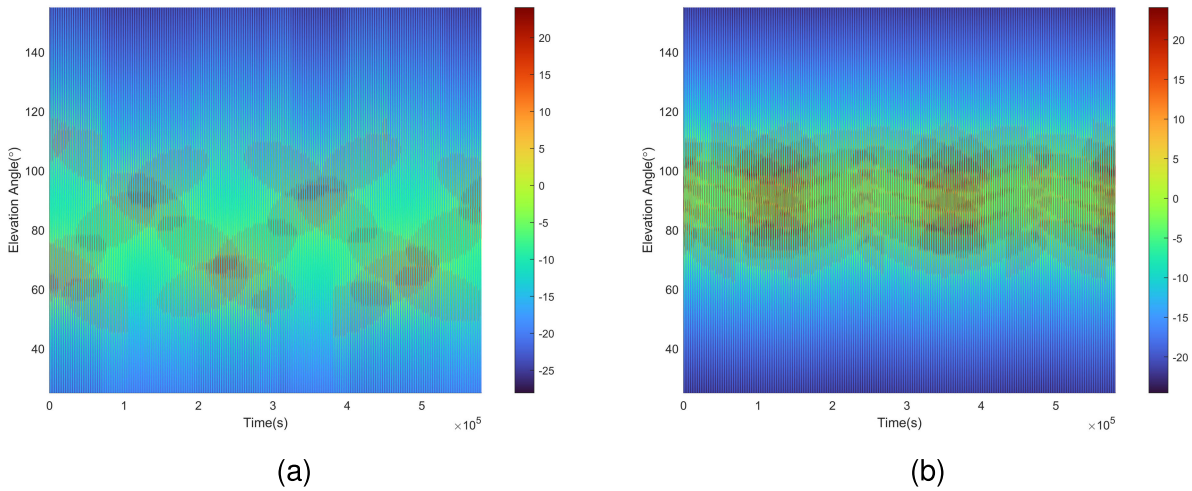

Fig. 6. The simulated TEIS for INR:(a) ground station latitude=10°,(b) ground station latitude =70°.

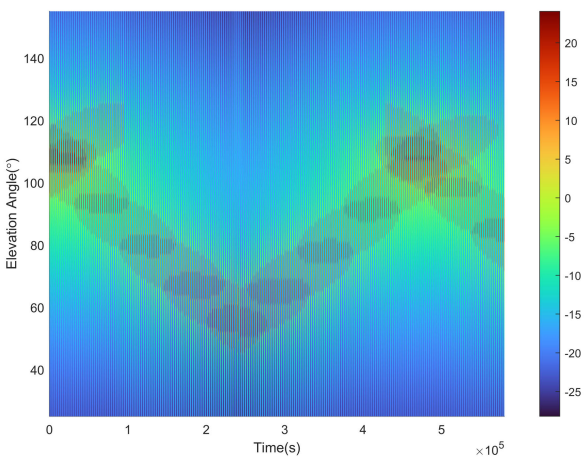

Fig. 7. The simulated TEIS for INR when the orbital plane number is 36.

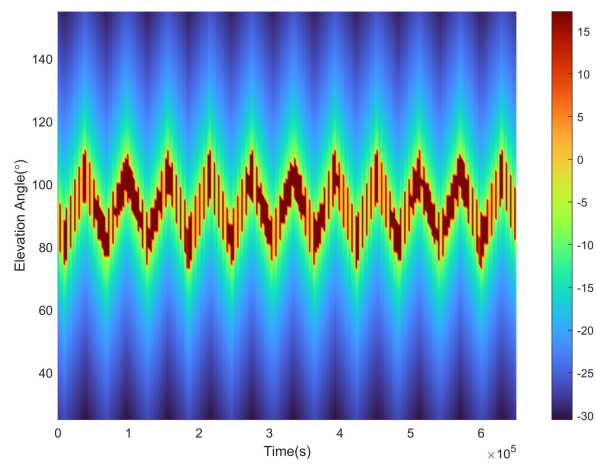

Fig. 8. The simulated TEIS for INR when the orbital height is 1200km.

angle domain. High-latitude ground terminals are more prone to sustained interference from satellites passing near the zenith.

Fig. 7 presents the simulated TEIS when the number of orbital planes is reduced from 72 to 36. It can be observed that during periods of heavy interference, the range of corresponding elevation angles becomes broader and the variation period increases. This is because the reduction in the number of orbital planes decreases both the number and spatial density of visible satellites for a ground terminal. As a result, the maximum elevation angle at which a satellite can be linked at a given time is reduced, leading to a wider distribution along the elevation axis. In addition, the decrease in the number of orbital planes increases the overall orbital period of the constellation, which in turn extends the periodicity of heavy interference patterns.

Fig. 8 shows the simulated TEIS when the orbital altitude increases from 550 km to 1200 km. It is evident that the periodicity of heavy interference becomes significantly shorter, while the elevation range of heavy interference concentrates more closely around 90°. Moreover, the "interference hotspots" become more densely overlapped, forming what can be described as an "interference line." This is attributed to the increase in orbital altitude, which introduces more interfering satellites with higher elevation angles. Additionally, the slower angular velocity of satellites at higher altitudes allows each satellite to remain within the field of view of the ground terminal for a longer duration. Consequently, each ground terminal is more likely to link with the same or adjacent satellites, reducing the spatial and temporal variability among interference hotspots. The shortened repetition period of heavy interference in Fig. 6 is also due to the increased number of interference sources and their prolonged visibility duration at higher altitudes, which slows the variation in satellite distribution within the ground terminal's visible domain. This makes it more likely for similar interference patterns to recur within shorter time intervals.

Fig. 9 compares the PDF of INR under different elevation angles when the orbital inclination is 80° and the interfered ground terminal is located in Beijing. Specifically, it examines

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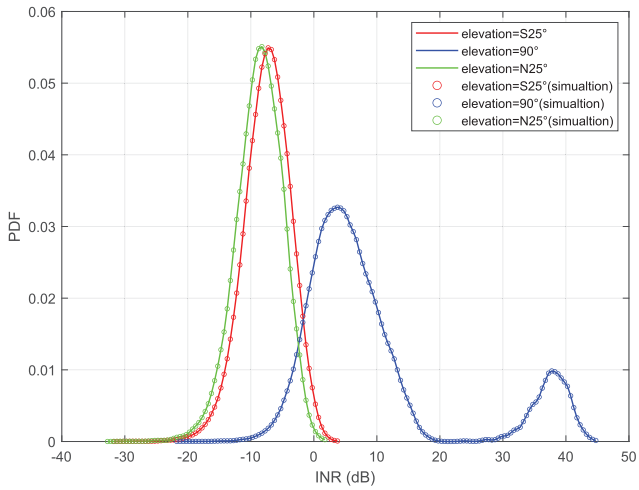

Fig. 9. Comparison of INR PDF at different elevation angles.

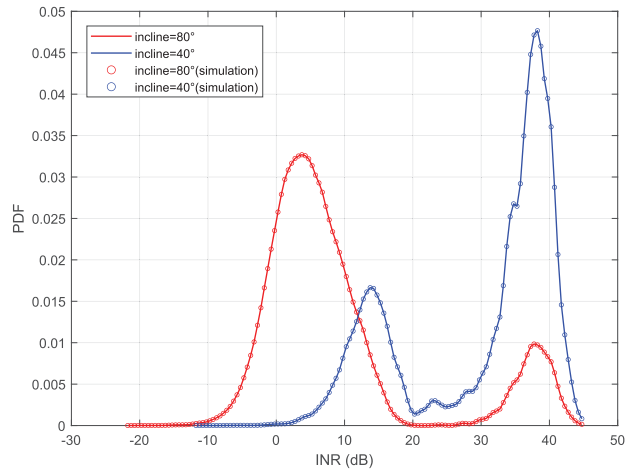

Fig. 10. Comparison of INR PDF at different orbital inclinations.

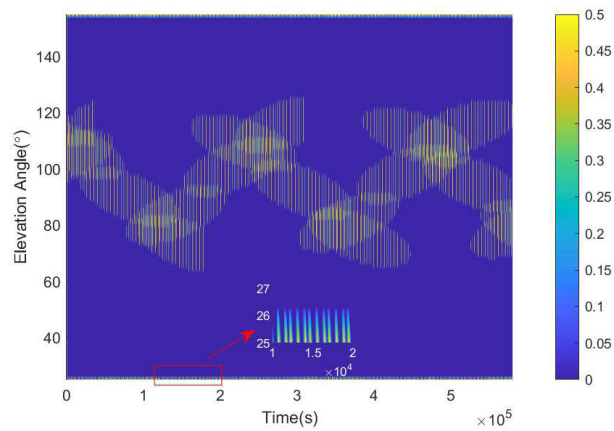

Fig. 11. The simulation result of outage probabilities.

three scenarios: the communication satellite is located at 25° south of the zenith, zenith, and 25° north of the zenith. The Monte Carlo simulations are used to validate the results. The simulation results show that when the communication satellite is located at 25° south or north, the interference intensity is relatively low, with almost INR values below 0 dB. This is because the elevation angle of the communication satellite is lower in these cases, while interfering ground terminals adopt a maximum-elevation-linking strategy. Consequently, the interfering satellites selected typically have high elevation angles and thus have limited ability to interfere with communication satellites at lower elevation angles.

Additionally, the interference when the communication satellite is located to the south is slightly higher than that of the one to the north. This is attributed to the lower latitude of the southern satellite, where the selected interfering satellites also tend to have lower elevation angles, resulting in slightly heavier interference. For the satellite at 90° elevation, the interference is significantly heavier and exhibits a "dual-peak" effect. This phenomenon arises because a larger number of interfering satellites are capable of causing interference at this elevation. Moreover, the higher the elevation angle of the interfering satellite, the shorter its duration from being selected for link establishment to link termination. As a result, the peak with heavier interference intensity appears lower than the peak with lower interference intensity.

Fig. 10 compares the INR PDFs for communication satellites at 90° elevation under different orbital inclinations. As shown in the figure, when the orbital inclination is reduced to 40°, the peak value corresponding to heavy interference becomes significantly higher, and even the low-interference intensity is higher than in the case of 80° inclination. This is because at 40° inclination, the ground terminal is located near the edge of the orbital coverage. In this situation, the field of view of the ground terminal contains a large number of high-elevation satellites. Furthermore, the reduction in inclination increases the satellite dwell time within the ground terminal's field of view, enabling more satellites to impose long-duration heavy interference on communication satellites at 90° elevation. Even in cases of weaker interference, the increased number of high-elevation interfering satellites leads to higher INR levels than those observed under higher orbital inclinations. Therefore, for a given ground terminal, changes in orbital inclination can significantly affect the distribution of interference intensity.

Fig. 11 presents the simulation results of outage probability when the orbital inclination is 80° and the ground terminals are located in Beijing. As shown in the figure, similar to Fig. 5 (a), the outage probability increases significantly within the "interference hotspot" regions where heavy interference occurs. This indicates a high likelihood of system outage within these time–elevation domains. Outside these regions, particularly in areas where only weak interference is observed in Fig. 5 (a), the outage probability remains nearly zero, suggesting that the system can tolerate a certain level of weak interference.

Notably, intermittently high outage probabilities appear near the edge elevation angles of the ground terminal's satellite visibility range. Although the interference intensity in these regions is relatively low, the received power of the communication signal is also low. As a result, the SINR occasionally falls below the system's required threshold, leading to temporary

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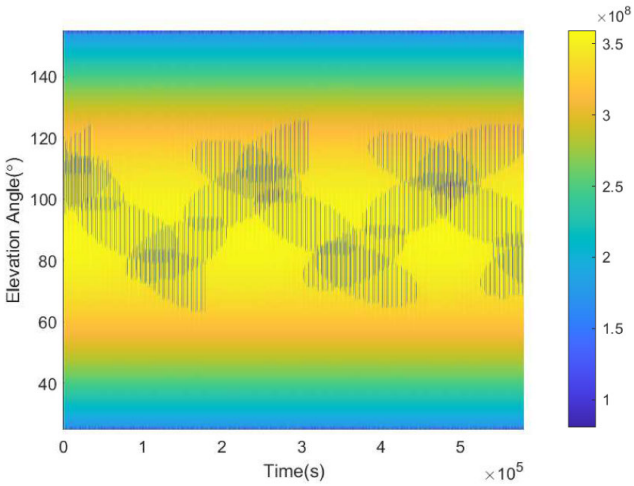

Fig. 12. The simulation result of throughput.

outages. Therefore, in interference avoidance strategies, attention should be paid not only to regions with heavy interference but also to low-elevation regions where the communication signal power is weak.

Fig. 12 depicts the simulation results of throughput when the orbital inclination is 80° and the ground terminals are located in Beijing. As shown in the results, the throughput gradually increases with the elevation angle. This is because higher-elevation links correspond to shorter propagation distances and higher SINR. However, in the high-elevation region where strong interference occurs, the throughput exhibits a noticeable decrease, indicating that co-frequency interference significantly degrades the transmission capability of the system. In addition, discrete low-throughput regions also appear at low elevation angles, which is consistent with the outage probability results. This is mainly due to the weaker received power at low elevations, making the system more susceptible to performance degradation.

V. Conclusion

Due to the challenges posed by the large number of interference sources, high interference dynamics, and complex interference link variations in NGSO mega-constellations, a time–elevation interference spectrum-based analysis method is proposed in this paper, which effectively overcomes the limitations of traditional approaches in capturing the macroscopic patterns of inter-constellation interference. By jointly considering the time and elevation domains, this method characterizes the temporal and spatial evolution of strong interference more comprehensively. In addition, a probabilistic model of inter-constellation interference is derived based on the joint spatial distribution of satellites and ground terminals, enabling further analysis of interference characteristics from a statistical perspective using the conditional elevation angle. To evaluate the impact of interference on system availability, the outage probability and throughput are also analyzed. Simulation results reveal that inter-constellation interference exhibits significant periodicity and hotspot concentration. Moreover, the interference characteristics in the time–elevation domain vary with constellation configuration parameters such as orbital inclination, number of orbital planes, orbital altitude, and ground terminal latitude. The proposed method and findings provide valuable insights for interference-aware constellation configuration design and ground station deployment.

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Zhaoyang Su received the B.E. degree in communication engineering from Beijing Jiaotong University (BJTU), Beijing, China, in 2020, where he is currently pursuing the Ph.D. degree in communication and information systems with the School of Electronics and Information Engineering, Institute of Broadband Wireless Mobile Communications. His research interests include satellite communications, high-speed railway communications, and V2V/V2l channel modeling.

Kai Wang received the B.E. degree in communication engineering from Beijing Jiaotong University (BJTU), Beijing, China, in 2018, where he is currently pursuing the Ph.D. degree in information and communication engineering with the School of Electronics and Information Engineering. His current research interests include channel measurement and modeling for high-speed railway communication systems.

Yuchen Cai received the B.E. degree in electronic information engineering from Zhengzhou University, Zhengzhou, China, in 2023. She is currently pursuing the Ph.D. degree in information and communication engineering with the School of Electronics and Information Engineering, Beijing Jiaotong University (BJTU). Her current research interests include vehicle networking communication technology and satellite communication technology.

Lipeng Ning has been engaged in the overall and implementation of satellite communication and navigation integrated communication projects for many years. He is currently a Senior Engineer with China Satellite Network Application Company Ltd. He is familiar with the characteristics and applicable scenarios of FDMA, TDMA, CDMA, 5G, and other communication systems. He has participated in a number of important projects throughout the process and has rich experience in engineering design, demonstration, and implementation.

Liu Liu (Member, IEEE) received the B.E. and Ph.D. degrees from Beijing Jiaotong University (BJTU), Beijing, China, in 2004 and 2010, respectively. He was a Post Ph.D. Researcher with the Institute of Broadband Wireless Mobile Communications, School of Electronics and Information Engineering, BJTU, from 2010 to 2012. Since 2018, he has been a Full Professor with the School of Electronics and Information Engineering, BJTU. Since 2021, he has also been the Vice-Principal of the School of Electronics and Information Engineering, BJTU. He has authored or co-authored more than 150 research papers in journals and conference proceedings. His current research interests include channel measurement and modeling for different propagation environments, signal processing of wireless communication in time-varying channels, next-generation mobile cellular systems, intelligent transport systems, vehicular communications, and satellite communications.

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Tao Zhou (Senior Member, IEEE) received the B.E. degree from Changchun University of Science and Technology, Changchun, China, in 2009, and the Ph.D. degree from Beijing Jiaotong University, Beijing, China, in 2016. From 2014 to 2015, he was a Visiting Ph.D. Student with the Centre for Communication Systems, School of Engineering and Computing Sciences, Durham University, U.K. Since 2022, he has been a Full Professor with the Institute of Broadband Wireless Mobile Communications, Beijing Jiaotong University. He has authored one academic book and more than 80 papers in refereed journals and conference proceedings. His current research interests include channel measurement and modeling, artificial intelligence for wireless communications, and B5G/6G key technologies. He is a Senior Member of Chinese Institute of Electronics and China Institute of Communications. He has served as a TPC Member for IEEE ICC 2020, IEEE VTC 2019 Fall, and IEEE ICCC 2021; the Session Chair for WCSP 2020; and the Track Chair for ICSPS 2022. He has been an Associate Editor of IEEE ACCESS, an Academic Editor of *International Journal of Antennas and Propagation*, a Guest Associate Editor of Radio Science Special Issue on Radio Channel Measurements and Modeling for Future High-Speed Railway Communications, and a Guest Editor of *Symmetry* Special Issue on Communication Engineering and Symmetry: Frontiers, Trends, and Challenges.

Bo Ai (Fellow, IEEE) received the master's and Ph.D. degrees from Xidian University, China. He received the honor of Excellent Post-Doctoral Research Fellow with Tsinghua University in 2007. He was a Visiting Professor with the Electrical Engineering Department, Stanford University, in 2015. He is currently a Full Professor with Beijing Jiaotong University, where he is the Dean of the School of Electronic and Information Engineering. He is one of the directors for Beijing "Urban Rail Operation Control System" International Science and Technology Cooperation Base and the Backbone Member of the Innovative Engineering-Based jointly granted by the Chinese Ministry of Education and the State Administration of Foreign Experts Affairs. He is the research team leader of 26 national projects. He has authored or co-authored eight books and authored more than 300 academic research articles in his research area. He holds 26 invention patents. Five articles have been the ESI highly-cited paper. He has been notified by the Council of Canadian Academies that based on the Scopus database, he has been listed as one of the top 1% authors in his field all over the world. He has also been feature interviewed by *IET Electronics Letters*. His research interests include the research and applications of channel measurement and channel modeling, and dedicated mobile communications for rail traffic systems. He is a fellow of the Institution of Engineering and Technology (IET Fellow) and the IEEE VTS Distinguished Lecturer. He received the Distinguished Youth Foundation and the Excellent Youth Foundation from the National Natural Science Foundation of China, the Qiushi Outstanding Youth Award by Hong Kong Qiushi Foundation, the New Century Talents by Chinese Ministry of Education, the Zhan Tianyou Railway Science and Technology Award by Chinese Ministry of Railways, and the Science and Technology New Star by Beijing Municipal Science and Technology Commission. He has also won some important scientific research prizes. He was the co-chair or a session/track chair of many international conferences. He is the IEEE VTS Beijing Chapter Vice Chair and the IEEE BTS Xi'an Chapter Chair. He is an Associate Editor of IEEE ANTENNAS AND WIRELESS PROPAGATION LETTERS and IEEE TRANSACTIONS ON CONSUMER ELECTRONICS and an Editorial Committee Member of the *Wireless Personal Communications* journal. He is a Lead Guest Editor of Special Issues on IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, IEEE ANTENNAS AND PROPAGATIONS LETTERS, and *International Journal on Antennas and Propagations*.

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